BANKHAUL · TRACEABILITY MATRIX

BankHaul traceability matrix

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DOCUMENT CONTROL

Document ID	BKH-RTM-001	Version	0.2
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/03b-traceability.md @ 659e1df		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

BankHaul traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 16 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-013	PR-001	No one in the water	BKH-CAL-001, BKH-BLD-001	Analysis of the build plan (TRL 3); field trial with partner fishers (TRL 4)	Met on paper
UR-005	PR-002	Working span	BKH-CAL-001, BKH-BLD-001	Analysis, BKH-CAL-001 A; measured at the trial site (TRL 4)	Met on paper: rope bought for 50 m
UR-004	PR-003	Hand pull	BKH-CAL-001, BKH-BLD-001, BKH-STM-001, BKH-OPT-001, BOM	Analysis, BKH-CAL-001 B; spring scale in trials (TRL 4)	Met on paper in still water: about 200 N (estimate); thin in a 0.3 m/s current (241 N) or on a stickier bank (243 N at friction 0.9); a smooth bank-edge board (BOM 18, O-2) is fitted and the R3 trial measures the pull with and without it
UR-008	PR-004	Bank anchor holding	BKH-CAL-001, BKH-BLD-001, BKH-STM-001	Analysis, BKH-CAL-001 D; pull test on a CalRig-style rig (TRL 4)	Met on paper: anchor factor 2.13 at 10 kPa
UR-008	PR-005	Far stake holding	BKH-CAL-001, BKH-BLD-001, BKH-STM-001	Analysis, BKH-CAL-001 E; pull test from the boat (TRL 4)	Met on paper: factor 2.77 at 10 kPa (stake yields first, D-S2), 2.03 at 5 kPa; at the top pin hole (1.2 m of water) 2.51 at 10 kPa, and the top hole is not used at 5 kPa (D-S7)
UR-007	PR-006	Cycle time	BKH-CAL-001, BKH-BLD-001, BKH-STM-001	Analysis, BKH-CAL-001 G; timed trials (TRL 4)	Met on paper: about 6.3 minutes (estimate)
UR-009	PR-007	Season durability	BKH-CAL-001, BKH-STM-001	Analysis, BKH-CAL-001 F; season pilot with an inspection log (TRL 5)	Met on paper (estimate): rope factor 5.7 after sun damage with both loop ends spliced (D-S6); UV loss, rope strength and bush wear plausible (STEM S18, S19, S22)
UR-011	PR-008	Cost per station (restated, decision 15A)	BKH-CAL-001	Bill of materials priced with the partner	Not met**: local-materials station about USD 98 (first-order estimate); steel trial station USD 399, accepted for TRL 4
UR-010	PR-009	Level change	BKH-CAL-001	Analysis, BKH-CAL-001 H; timed relocation (TRL 4)	Met on paper: about 22 minutes (estimate); five pin holes at 200 mm
UR-006	PR-010	Weak link			

User requirement	Product requirement	Title	Evidence	Verification	Status
			BKH-CAL-001, BKH-BLD-001	Analysis, BKH-CAL-001 C; pull three samples of each cord batch (TRL 4)	Met on paper; cord chosen by test when bought (STEM S6)
UR-003	PR-011	Working distance	BKH-CAL-001, BKH-BLD-001, BKH-STM-001	Inspection of the layout (TRL 3); measured at the trial site (TRL 4)	Met by layout: 5.0 m, as the Be Crocwise advice asks (STEM S24)
UR-002	PR-012	Own nets	BKH-CAL-001	Inspection of the model and build plan (TRL 3); trials with partner nets (TRL 4)	Met by design
UR-012	PR-013	Local making	BKH-CAL-001	Inspection of the BOM and build plan (TRL 3)	Met by design
UR-013	PR-014	Two-person lift	BKH-CAL-001, BKH-STM-001	Analysis, BKH-CAL-001 J	Met on paper with the lowering rope (BKH-DEC-001 O-7, decided by Amish 2026-10-25): far stake 33.2 kg, two people, 16.6 kg each; the stake is stood up against the boat side and let down on a rope through its top cross hole, so no one holds it at arm's length over the side; to confirm in the TRL 4 drill
UR-008	PR-015	Pin plates	BKH-CAL-001	Analysis, BKH-CAL-001 D	Met on paper: least ratio 1.15 (pad eye, D-S1)
UR-014	PR-016	Authority agreement	BKH-CAL-001, BKH-STM-001	Demonstration with the co-design partner (TRL 4)	To verify at TRL 4

Gaps

None.